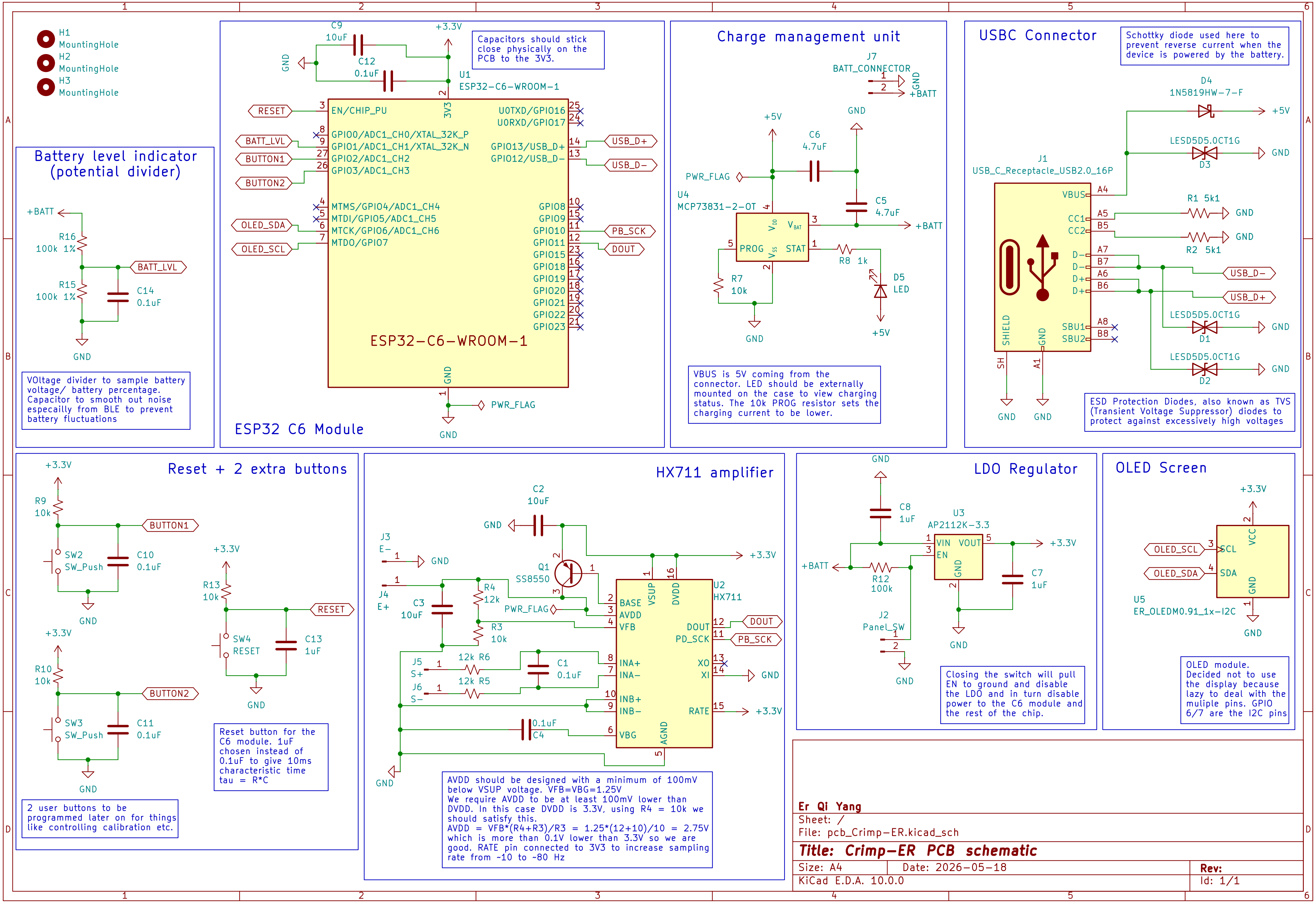
H1 MountingHole
H2 MountingHole
H3 MountingHole
Battery level indicator (potential divider)
+BATT
R16 100k 1%
R15 100k 1%
C14 0.1uF
BATT_LVL
GND
VOltage divider to sample battery voltage/ battery percentage. Capacitor to smooth out noise especailly from BLE to prevent battery fluctuations
C9 10uF
C12 0.1uF
+3.3V
GND
Capacitors should stick close physically on the PCB to the 3V3.
U1
ESP32-C6-WROOM-1
RESET
BATT_LVL
BUTTON1
BUTTON2
OLED_SDA
OLED_SCL
EN/CHIP_PU
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
MTMS/GPIO4/ADC1_CH4
MTDI/GPIO5/ADC1_CH5
MTCK/GPIO6/ADC1_CH6
MTDO/GPIO7
3V3
U0TXD/GPIO16
U0RXD/GPIO17
GPIO13/USB_D+
GPIO12/USB_D-
GPIO8
GPIO9
GPIO10
GPIO11
GPIO15
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
USB_D+
USB_D-
PB_SCK
DOUT
ESP32-C6-WROOM-1
GND
PWR_FLAG
ESP32 C6 Module
Charge management unit
J7
BATT_CONNECTOR
GND
+BATT
+5V
C6 4.7uF
C5 4.7uF
PWR_FLAG
U4
MCP73831-2-OT
V_{DD}
V_{BAT}
PROG
V_{SS}
STAT
R8 1k
R7 10k
D5 LED
+5V
VBUS is 5V coming from the connector. LED should be externally mounted on the case to view charging status. The 10k PROG resistor sets the charging current to be lower.
USBC Connector
Schottky diode used here to prevent reverse current when the device is powered by the battery.
D4
1N5819HW-7-F
+5V
LESD5D5.0CT1G
D3
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
R1 5k1
R2 5k1
USB_D-
USB_D+
LESD5D5.0CT1G
D1
LESD5D5.0CT1G
D2
SH
A1
ESD Protection Diodes, also known as TVS (Transient Voltage Suppressor) diodes to protect against excessively high voltages
Reset + 2 extra buttons
+3.3V
R9 10k
BUTTON1
SW2 SW_Push
C10 0.1uF
R10 10k
BUTTON2
SW3 SW_Push
C11 0.1uF
R13 10k
RESET
SW4 RESET
C13 1uF
2 user buttons to be programmed later on for things like controlling calibration etc.
Reset button for the C6 module. 1uF chosen instead of 0.1uF to give 10ms characteristic time tau = R*C
HX711 amplifier
C2 10uF
J3 E-
J4 E+
Q1 SS8550
C3 10uF
R4 12k
R3 10k
PWR_FLAG
J5 S+
J6 S-
12k R6
12k R5
C1 0.1uF
0.1uF C4
U2
HX711
BASE
AVDD
VFB
INA+
INA-
INB+
INB-
VBG
VSUP
DVDD
AGND
DOUT
PD_SCK
XO
XI
RATE
DOUT
PB_SCK
+3.3V
GND
AVDD should be designed with a minimum of 100mV below VSUP voltage. VFB=VBG=1.25V
We require AVDD to be at least 100mV lower than DVDD. In this case DVDD is 3.3V, using R4 = 10k we should satisfy this.
AVDD = VFB*(R4+R3)/R3 = 1.25*(12+10)/10 = 2.75V which is more than 0.1V lower than 3.3V so we are good. RATE pin connected to 3V3 to increase sampling rate from ~10 to ~80 Hz
LDO Regulator
C8 1uF
U3
AP2112K-3.3
VIN
VOUT
EN
GND
+3.3V
+BATT
R12 100k
C7 1uF
J2
Panel_SW
Closing the switch will pull EN to ground and disable the LDO and in turn disable power to the C6 module and the rest of the chip.
OLED Screen
+3.3V
OLED_SCL
OLED_SDA
SCL
SDA
VCC
GND
U5
ER_OLEDM0.91_1x-I2C
OLED module. Decided not to use the display because lazy to deal with the muliple pins. GPIO 6/7 are the I2C pins
Er Qi Yang
Sheet: /
File: pcb_Crimp-ER.kicad_sch
Title: Crimp-ER PCB schematic
Size: A4
Date: 2026-05-18
KiCad E.D.A. 10.0.0
Rev:
Id: 1/1